```
.options vntol=1m abstol=1u
.param tdv=1 tdt=1 fn1=50
.param omega_n={2*pi*fn1}

.tran 0 1000ms 0 0.1ms

U1Nenn=115k FREQU=50 Texp=5m TD=0
```

psir=1 psis=-0.5 psit=-0.5 (default)

rkessel=1e5
RUYy0_110k_10k_40MVA

t_Start=40m

normal inrush

PhaseA1

Trafo1

.param R=1 L={3/314.159}

High R -> faster swing in

psi_0=0 tpsi0=40m

Psi is only dependant of U and N1(windings)

Psi = Phi*N1
Psi=U(phase_peak)/omega

N1=1000

U=115000*sqrt(2)/sqrt(3)
=93883V (Peakvalue)

PhiN=U/omega/N1=
93883/314.159/1000
=0.2988 (Peakvalue)

V(Trafo1:u:w1:phi)
V(Trafo1:v:w1:phi)
V(Trafo1:w:w1:phi)

t_Start=40m

inrush with full pos. remanent flux

PhaseA2

Trafo2

psi_0=1 tpsi0=40m

t_Start=40m

inrush with full neg. remanent flux

PhaseA3

Trafo3

psi_0=-1 tpsi0=40m

t_Start=45ms

Inrush with full pos. remanent flux
(but at switching timepoint 45ms)

PhaseA4

Trafo4

psi_0=1 tpsi0=45m
psir=0 psis=0.866,psit=-0.866